

Distributed Systems

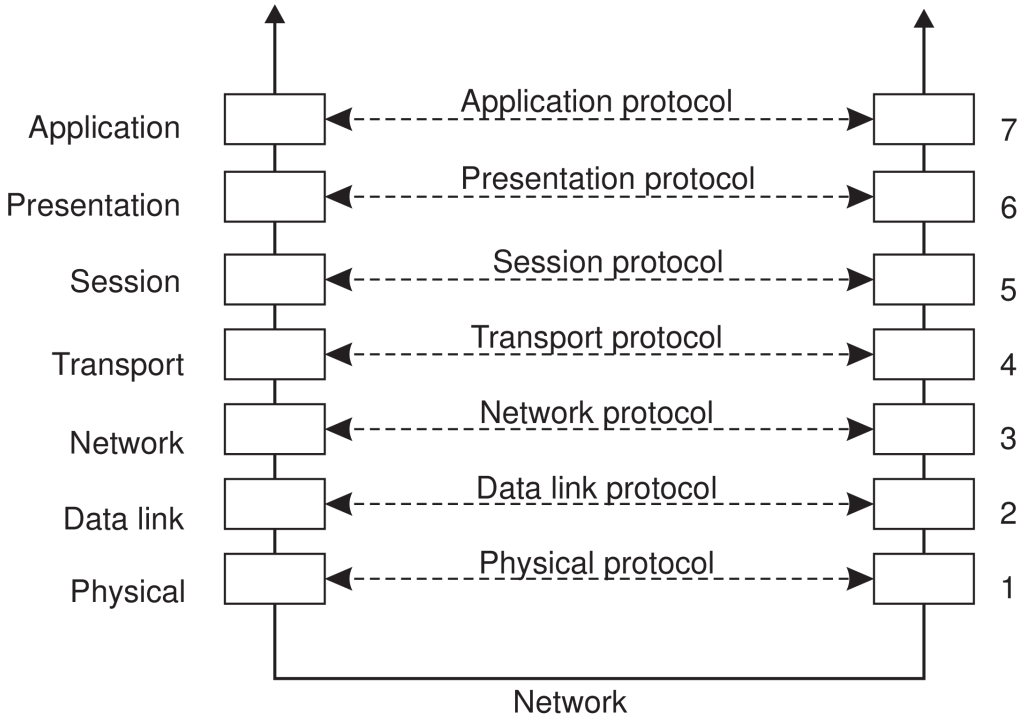
Lecture13: Communication

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Basic Networking Model



- **Drawbacks:**
  - Focus on message-passing only
  - Often unneeded or unwanted functionality
  - Violates access transparency

Low-level Layers

- **Recap:**
  - **Physical layer:** contains the specification and implementation of bits, and their transmission between sender and receiver
  - **Data link layer:** prescribes the transmission of a series of bits into a frame to allow for error and flow control
  - **Network layer:** describes how packets in a network of computers are to be routed.
- **Observation:**
  - For many distributed systems, the lowest-level interface is that of the network layer.

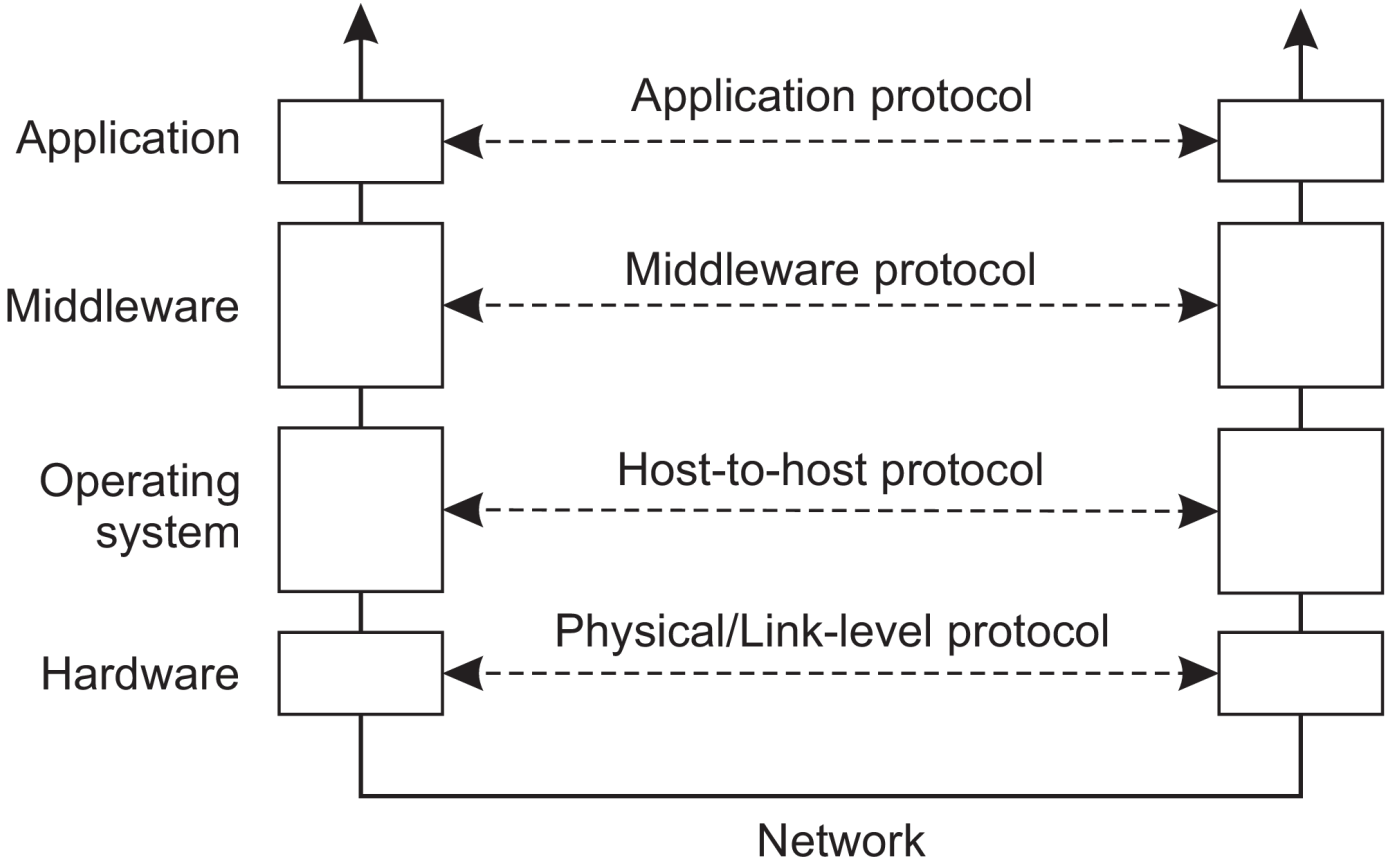
Transport Layer

- The transport layer provides the actual communication facilities for most distributed systems.
- **Standard Internet protocols:**
  - TCP: connection-oriented, reliable, stream-oriented communication
  - UDP: unreliable (best-effort) datagram communication

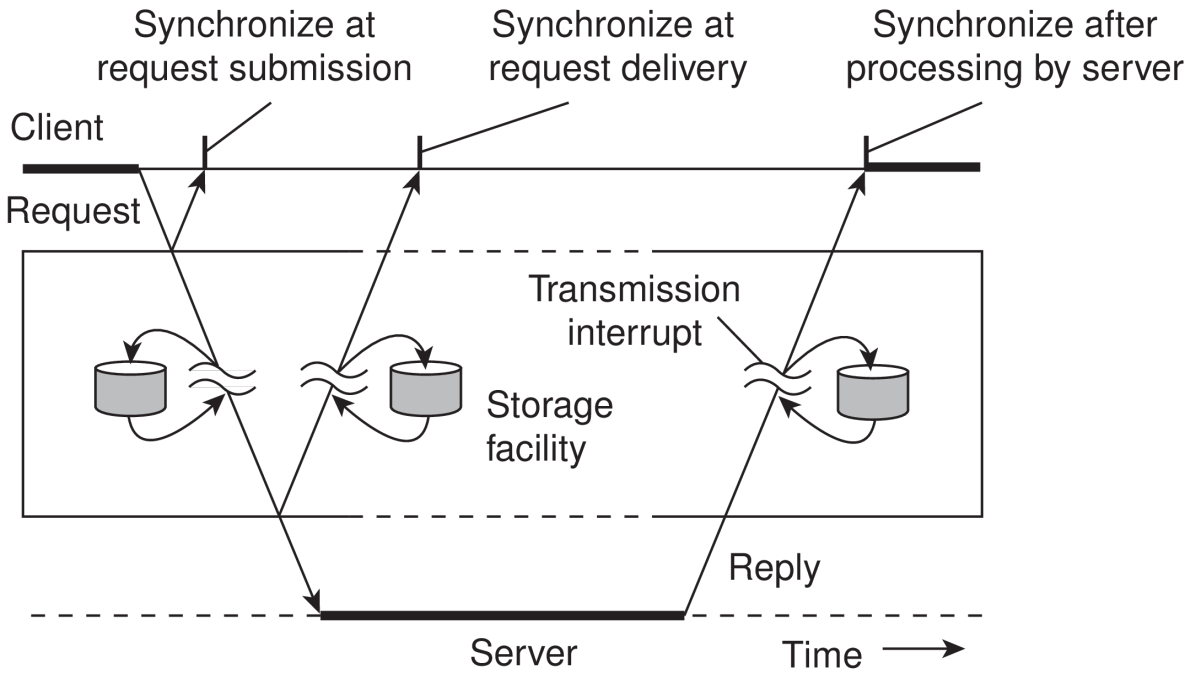
Middleware layer

- Middleware is invented to provide common services and protocols that can be used by many different applications.
  - A rich set of communication protocols
  - (Un)marshaling of data, necessary for integrated systems
  - Naming protocols, to allow easy sharing of resources
  - Security protocols for secure communication
  - Scaling mechanisms, such as for replication and caching

An Adapted Layering Scheme

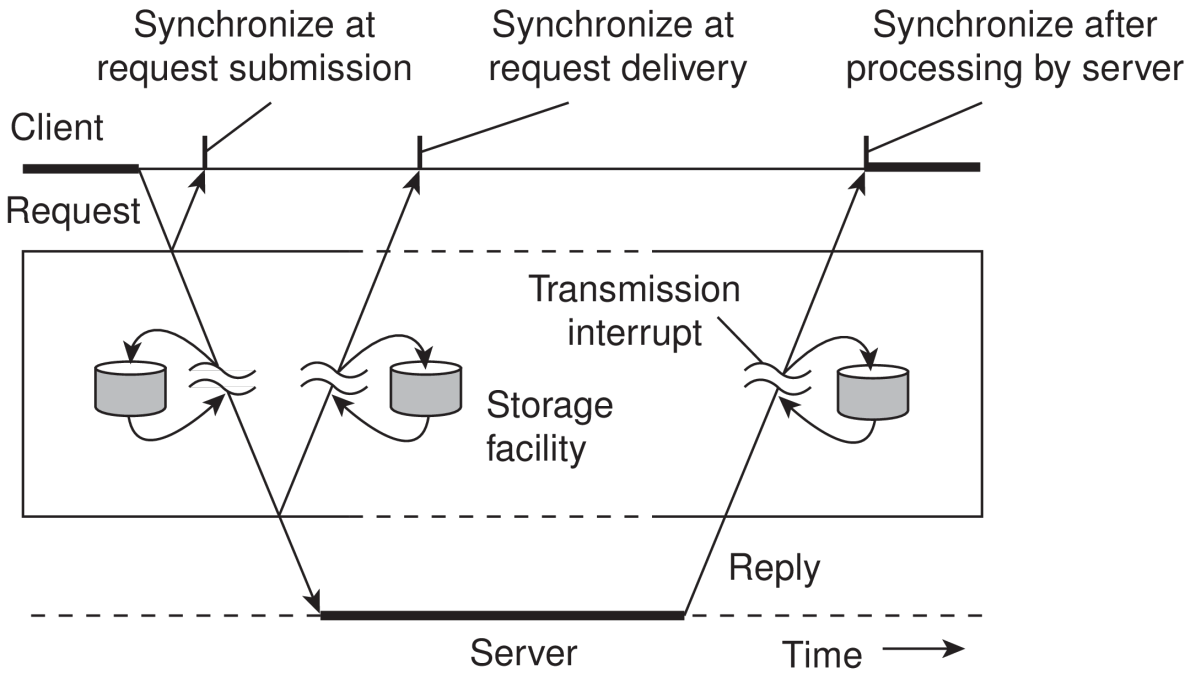


Types of Communication



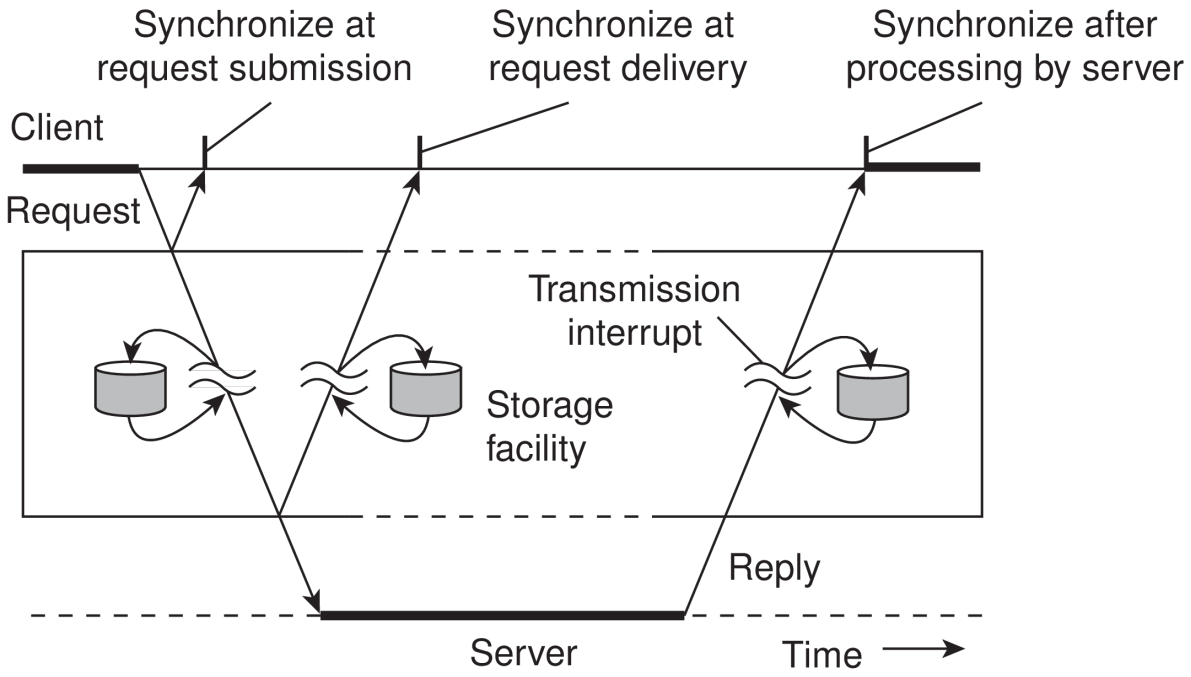
- Transient versus persistent communication
- Asynchronous versus synchronous communication

Types of Communication



- **Transient communication:** Comm. server discards message when it cannot be delivered at the next server, or at the receiver.
- **Persistent communication:** A message is stored at a communication server as long as it takes to deliver it.

Types of Communication



- At request submission
- At request delivery
- After request processing

Client / Server

- Client/Server computing is generally based on a model of transient
- **Synchronous Communication:**
  - Client and server have to be active at the time of communication
  - Client issues request and blocks until it receives reply
  - Server essentially waits only for incoming requests, and subsequently processes them.
- **Drawbacks synchronous communication:**
  - Client cannot do any other work while waiting for reply
  - Failures have to be handled immediately: the client is waiting
  - The model may simply not be appropriate (mail, news)

Messaging

- **Message-oriented middleware:**
- Aims at high-level persistent asynchronous communication:
  - Processes send each other messages, which are queued
  - Sender need not wait for immediate reply, but can do other things
  - Middleware often ensures fault tolerance

Wrap-Up + What's Next

In the next lecture, we will discuss the Remote Procedure Call topic, including:

- Basic RPC,
- Parameter Passing,
- RPC-based Application,
- Variations on RPC.